

For every sensor, we can compute two indicators (at least). They're called α_i and β_i for the generic sensor i . The former takes the name of minimum sparsity attack on i (a minimum sparsity attack is one targeting the smallest number of points in the system). Assume we can attack meter i , then how many additional meters do we need to attack an strong FDIA attack?

$$\begin{aligned} \forall i \quad \alpha_i = \min \|a\|_0 \\ \text{subject to} \\ a = Hc \quad \text{strong FDIA condition} \\ a_i = 1 \quad \text{injection of 1 unit of bad} \\ \text{data into meter } i \end{aligned}$$

Meters with small α_i are weak points.

Example

$$\alpha_3 = 6.$$

This means that if we control meter 3, then we need to take control of additional 5 (6-1) meters.

This information is very useful when organising protection against FDIA.



We have that (K^*, α_i) refer to minimum sparsity attacks.

The second local indicator is β_i . This is linked with the minimum magnitude.

$$\begin{aligned} \forall i \quad \beta_i = \min \|a\|_1 \\ \text{subject to} \\ a = Hc \quad \text{strong FDIA condition} \\ a_i = 1 \quad \text{injection of 1 unit of bad} \\ \text{data into meter } i \end{aligned}$$

We have that:

$$\|v\|_1 = \sum_{i=1}^{n_v} |v_i| \quad \text{ex: } \left\| \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} \right\|_1 = 6$$

So, $\|a\|_1$ is the sum of the magnitudes of the components of a .

β_i looks for strong FDIA targeting i and injecting the minimum amount of false data.

We can then divide between random and targeted FDIA. When we talk about random, we mean that we inject $a = Hc$ with a random choice for c . For targeted, it means that we want to modify $\hat{x}$ in a specific way. Remember that $\hat{x}_a = \hat{x} + c$. We choose a vector c that produces desired modifications of the estimate $\hat{x}$.

Now, let's discuss protection against FDIA.

We have two protection measures:

1. We can secure (/ protect) some sensors. Call $P \subseteq \{1, 2, \dots, m\}$ the set of protected meters. Meters in P cannot be attacked.
2. Directly measure some state variables, this is done with devices known as PMU (phasor measurement units). They can directly measure voltage. The problem with these sensors is that they need to use GPS or similar data, having some vulnerabilities. Call $Q \subseteq \{1, 2, \dots, n\}$ the set of state variables directly measured with PMU.

How to best decide P and Q , given that we have a budget for protection? We have to first write the FDIA problem when we consider protection measures.

The condition is always $a = Hc$ (strong FDIA), with $c \neq 0$ (non trivial attack), and $a_i = 0 \forall i \notin K$ (attacker cannot corrupt the meters outside the attack set).

If a meter is protected, we add a new condition given by $a_i = 0 \forall i \in P$, for which protected meters cannot be attacked. Also, some state variables are directly measured, so $c_i = 0 \forall i \in Q$.

From the defender perspective, we want to solve the problem of efficient allocation of protections. What is the best allocation of P and Q protection measures?

We have a specific budget, and we have to decide whether to reinforce meters (P) or we buy PMU to measure state variables (Q).

One idea is to increase as much as possible the difficulty (# of meters to attack) of the simplest attack. It's basically a max-min problem, trying to model the defender/attacker interaction.

$$\max_{(P,Q)} = \{ \min_{i=1,2,\dots} \alpha_i \}$$

subject to

$$\alpha_i = \min \|a\|_0$$

subject to

$$a = Hc$$

$$\alpha_i = 1$$

$$a_i = 0 \quad \forall i \in P$$

$$c_i = 0 \quad \forall i \in Q$$

$$-\text{cost}(P, Q) = \text{budget}$$

We have two optimization problems nested one in other.

One note: the problem of $\alpha_i = \min \|a\|_0$ subject to $a = Hc$ and $a_i = 1$ is a non trivial problem, while the problem of finding $\hat{x}$ such that it's close as possible to the real x is easier. The function inside that problem, $J(\cdot)$, is a continuous function in x , and this makes possible to compute derivatives, finding points where the derivative is zero. We can then find the min by computing the derivative and put it equal to zero.

In the other hand, the above problem is different. While a is still a continuous variable, the target function (being the zero norm of a , $\|a\|_0$), what is it? *Reflect on it at home, is it a continuous function, and so can we compute derivatives on it?*